

Representation Matters: Cross-Dataset Analysis of Implicit Hate Speech Detection Methods

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Abstract

Implicit hate speech expresses prejudice indirectly, using nuanced or sarcastic language instead of overtly offensive words. Detecting such language requires models that capture semantic and contextual meaning. This project explores how different embedding strategies affect classification performance when applied to implicit hate detection. We will experiment with three models—Shared Contrastive Learning (SharedCon), BiLSTM, and Transformer Fine-Tuning—to evaluate which model best generalizes across datasets.

1 Introduction

Hate speech comprises language that insults, threatens, or demeans people on the basis of identity (e.g. race, religion, gender, sexual orientation). As discourse shifts online, communication is faster and more public, making harmful content more visible and consequential. Recent work reports that nearly 30% of young people have experienced some form of cyberbullying, illustrating the urgency of developing effective hate speech detection systems [7].

Detecting hate speech is fundamentally an NLP task, requiring models to analyze wording, context, tone, and implicit meaning. This is especially challenging in the case of implicit hate speech, where hostility is conveyed through subtle phrasing, metaphors, or coded references rather than explicit slurs. These nuances cause significant inconsistencies across datasets, and many existing NLP models perform well only within a single dataset and fail to generalize across others [13].

To address this, our project will compare three NLP strategies that differ in how they represent linguistic meaning: (1) a contrastive model based on word embeddings, (2) a target-aware classification model, and (3) a transformer model fine-tuned with adversarial training. Comparing these approaches across multiple datasets will help evaluate how representation choices affect both classification performance and cross-dataset generalization.

2 Motivation

Implicit hate speech is one of the most important challenges in the realm of online safety. Explicit hate speech, which uses identifiable hateful slurs, is different from implicit hate speech because the latter uses subtle language techniques, which generally cannot be detected by the keywords of human moderators because implicit hate speech is mostly made up of innocent-looking phrases, but 33% [5] of the hateful content on the Web is implicit, according to some recent studies.

However, the current state-of-the-art solutions, including those in the work represented by SharedCon, face material challenges.

The most prominent is the clear deterioration of their processing efficiency on other sources of data, apart from the one they are originally trained on, marking them with poor adaptability or lack of quality to be generally applicable. The continuously developing nature of hate speech, with its constantly developing coded vocabulary, hence linguistic techniques, calls for adaptability on the part of the solution proposed here.

3 Datasets

We use three publicly available hate-speech datasets in our experiments, summarized in Table 1. IHC and SBIC include annotations for implied meaning and target groups, while DynaHate focuses solely on surface-level hate expressions without implied context.

- **IHC [5]:** A hate-speech dataset collected from Twitter hate communities and their followers. It contains both the *target* and *implied meaning* of each post, with 6,346 implicit hate samples out of 22,056 total tweets.
- **SBIC [11]:** An offensive-language dataset with hierarchical annotations capturing *offensiveness*, *target*, and *implied statements*, designed to study social bias and stereotypical implications.
- **DynaHate [12]:** A dynamically generated hate-speech dataset created using a combination of human-written and model-generated examples to evaluate model robustness and deception resistance.

Table 1: The statistical information of the three datasets to be used in our experiments.

Dataset	Train Set	Validation Set	Test Set
IHC	12,234	4,078	4,079
SBIC	35,424	4,666	4,691
DynaHate	32,924	4,100	4,120
TOTAL	80,582	12,844	12,890

4 Examples from Dataset

Content Warning: The following examples are taken from publicly available hate-speech datasets and may contain offensive or upsetting language. They are included strictly for academic and research purposes.

Each dataset includes structured annotations enabling analysis of both explicit and implicit bias.

- **IHC** provides variables such as *class*, *implicit subtype*, and *implied statement*.
- **SBIC** includes *offensiveYN*, *targetMinority*, and *hasBiasedImplication*.

- **DynaHate** contains *text*, *label*, and *target*.

Table 2 shows representative examples from the three datasets—Implicit Hate Corpus (IHC), Social Bias Inference Corpus (SBIC), and DynaHate—highlighting their diversity in hate expression.

Table 2: Representative samples from IHC, SBIC, and DynaHate datasets.

Text (shortened)	Label	Dataset
“The white race is finished.”	implicit_hate	IHC
“Imagine if the media had paid attention...”	not_hate	IHC
“Every half-breed cop in America is trying to rape white women.”	offensive	SBIC
“Somebody toss this girl a Dasani or something, she’s #THIRSTY.”	not_offensive	SBIC
“Dalits are lowlives.”	hate	DynaHate
“Asian people are the shit.”	not_hate	DynaHate

5 Data Preprocessing

We conduct our experiments on three publicly available hate speech detection datasets, each capturing different aspects of online hate speech. Table 1 presents the statistical distribution of samples across training, validation, and test sets for all three datasets.

5.1 Dataset Preparation

Implicit Hate Corpus (IHC) [5]: The IHC dataset focuses on implicit hate speech detection, where hateful intent is expressed indirectly without explicit slurs or offensive language. We use the Stage 1 posts file (`implicit_hate_v1_stg1_posts.tsv`) containing text-label pairs. Following the dataset’s design, we retain only two classes:

- *implicit_hate*: Posts containing implicit hateful content
- *not_hate*: Non-hateful posts

We exclude the *explicit_hate* class to focus specifically on implicit hate detection, which presents a more challenging classification task. The dataset is split into training (60%), validation (20%), and test (20%) sets using stratified sampling to maintain class balance, resulting in 12,234 training samples, 4,078 validation samples, and 4,079 test samples.

Social Bias Inference Corpus (SBIC) [11]: SBIC provides annotated social media posts with crowdsourced labels indicating offensive content and bias. We utilize the aggregated files `SBIC.v2.agg. {trn/dev/tst}.csv` where each unique post has a single aggregated label based on multiple annotator judgments. The offensive score (`offensiveYN`) represents the fraction of annotators who marked a post as offensive. We convert this continuous score to binary labels:

- *offensive*: Posts where `offensiveYN > 0.5`
- *not_offensive*: Posts where `offensiveYN ≤ 0.5`

The dataset provides pre-defined splits with 35,424 training samples, 4,666 validation samples, and 4,691 test samples, which we use directly to ensure consistency with prior work.

DynaHate [12]: DynaHate is a dynamically generated dataset created through an adversarial human-and-model-in-the-loop approach. The dataset includes both original and augmented examples designed to challenge hate speech classifiers with difficult cases. We use all available rounds (R1–R4) of data generation to maximize dataset size and diversity. The dataset contains binary labels:

- *hate*: Hateful content
- *nothate*: Non-hateful content

Following the original split configuration provided in the dataset, we obtain 32,924 training samples, 4,100 validation samples, and 4,120 test samples.

5.2 Text Preprocessing

Social media text presents unique challenges due to its informal nature, including URLs, user mentions, hashtags, and inconsistent formatting. We apply a standardized preprocessing pipeline to all datasets to normalize text while preserving semantic content.

Preprocessing Steps:

- **URL Removal:** We remove all URLs (beginning with `http://`, `https://`, or `www.`) as they typically do not contribute to hate speech classification and may introduce noise.
- **Mention Removal:** User mentions (strings beginning with `@`) are removed to anonymize content and reduce vocabulary sparsity.
- **Hashtag Normalization:** Hashtags are converted to plain text by removing the `#` symbol while retaining the word (e.g., `#hate` → `hate`). This preserves the semantic content while normalizing the format.
- **Whitespace Cleaning:** Multiple consecutive spaces are collapsed into single spaces, and leading/trailing whitespace is removed.

These preprocessing steps are consistently applied across all three datasets before clustering and training to ensure fair comparison. The preprocessing function handles missing or non-string values by returning empty strings, ensuring robust data pipeline execution.

Rationale: This minimal preprocessing approach balances text normalization with semantic preservation. Unlike aggressive preprocessing that removes stop words or applies stemming, our approach maintains the original text structure, which is particularly important for contextual embeddings like HateBERT that benefit from complete sentence context. The preprocessing primarily focuses on removing social media-specific noise elements that are not informative for hate speech detection while preserving linguistic patterns that may indicate implicit hate.

All preprocessing is implemented using Python’s `re` (regular expressions) module and applied uniformly before any downstream processing, including clustering and model training. This ensures consistency across both embedding approaches (FastText and HateBERT) and all experimental configurations.

6 Methodology

6.1 Overview

To address the challenge of implicit hate speech detection, we implement and evaluate three distinct deep learning approaches that

leverage different architectural paradigms and training strategies. Each approach aims to capture the nuanced linguistic patterns that characterize implicit hate speech, where harmful intent is expressed through indirect language, coded references, or contextual implications rather than explicit slurs or offensive terms.

Approach 1 adapts the SharedCon framework [1], combining word-level embeddings with supervised contrastive learning to learn discriminative representations that separate hate speech from benign content while maintaining semantic coherence within each class.

Approach 2 employs a target-aware classification strategy that explicitly models the relationship between hateful content and its targets, incorporating target entity information to improve detection of context-dependent implicit hate.

Approach 3 leverages transformer-based architectures with adversarial training, applying perturbation-based regularization to enhance model robustness and improve generalization across diverse datasets and linguistic styles.

All three approaches are trained and evaluated on three benchmark datasets—IHC [5], SBIC [11], and DynaHate [12]—using both in-dataset and cross-dataset validation protocols to comprehensively assess performance and generalization capability. The following subsections provide detailed descriptions of each approach’s methodology, architecture, experimental setup, and training procedures.

6.2 Evaluation Protocol

We apply a unified evaluation protocol across all three approaches to ensure comparability in both within-dataset performance and cross-dataset generalization.

6.2.1 In-Domain Evaluation. In-domain evaluation trains and tests on the same dataset (e.g., IHC→IHC, SBIC→SBIC, DynaHate→DynaHate). Following the original dataset splits, we merge the training and validation sets ($\approx 80\%$ of the data) and reserve the original test set ($\approx 20\%$) for final evaluation.

6.2.2 Cross-Domain Evaluation. Cross-domain evaluation trains on one dataset and tests on another (e.g., IHC→SBIC, SBIC→DynaHate). We evaluate all nine train–test combinations across the three datasets.

6.2.3 Evaluation Metrics. We report two complementary metrics:

- **Macro-F1:** Equally weights both classes and is well suited for imbalanced datasets.
- **AUPRC:** Measures ranking ability for the positive class and provides a more informative evaluation than ROC-AUC in skewed binary classification.

6.2.4 Confusion Matrix Analysis. For each dataset’s in-domain results, we generate confusion matrices for the best-performing model (based on Macro-F1). These matrices highlight false positives, false negatives, and class-specific performance trends, offering insight into common failure modes and guiding future improvements.

6.3 Approach 1: Word Embedding + SharedCon Adaptation

Our first approach adapts the SharedCon framework [1] to work with word-level embeddings, leveraging both static and contextualized representations to capture implicit hate speech patterns. Unlike the original SharedCon which uses sentence-level embeddings, our adaptation focuses on token-level features to better capture subtle linguistic cues.

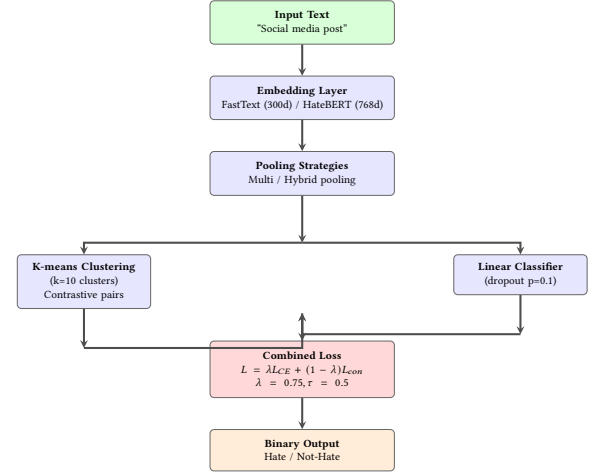


Figure 1: Architecture of Approach 1 (Word Embedding + SharedCon Adaptation). The pipeline processes input text through embedding (FastText or HateBERT), applies pooling strategies, performs K-means clustering for contrastive pairs, and trains with a combined loss integrating cross-entropy and supervised contrastive learning.

6.3.1 Experimental Setup. We evaluate on the three datasets described in Section 5 using Macro-F1 and AUPRC metrics, since these handle class imbalance effectively and provide threshold-independent evaluation [4]. Cross-dataset validation assesses generalization by training on one dataset and testing on others.

Our configuration uses NVIDIA GPU with Python 3.8, PyTorch 1.12 [10], and transformers 4.21 [14]. Training parameters include learning rates of $1e-5$ (FastText) and $2e-5$ (HateBERT), batch size 8, 5 epochs, $\lambda = 0.75$, and $\tau = 0.5$, with early stopping based on validation F1.

6.3.2 Model Architecture. We implement two embedding approaches:

FastText Encoder: Uses pre-trained 300-dimensional static embeddings [2] with character n-gram subword information, effectively handling out-of-vocabulary words common in social media hate speech. Multi-pooling operations (max, average, minimum) across tokens create fixed-size representations capturing different semantic aspects.

HateBERT Encoder: Employs domain-specific HateBERT [3] for 768-dimensional contextualized embeddings. We implement multi-pooling (max/avg/min operations) and hybrid-pooling (multi-pooling combined with CLS token representation).

Both encoders connect to a linear classification head with dropout ($p = 0.1$). To support SharedCon, we perform K-means clustering [9] ($k = 10$ per class) on training embeddings, creating diverse positive and negative pairs for contrastive learning.

6.3.3 Training Procedure. SharedCon Framework: The SharedCon framework [1] leverages shared semantic representations for implicit hate detection. It addresses coded language and contextual cues by learning shared patterns between semantically similar instances. Our adaptation maintains this principle at the word-embedding level.

The model optimizes a combined objective:

$$L_{\text{combined}} = \lambda \times L_{\text{CE}} + (1 - \lambda) \times L_{\text{con}} \quad (1)$$

where L_{CE} is cross-entropy loss, L_{con} is supervised contrastive loss [6] with temperature scaling ($\tau = 0.5$), and λ controls the objective balance.

Contrastive Learning: The supervised contrastive component creates structured embedding spaces where same-class samples cluster together while different classes maintain large margins. For implicit hate speech, this groups subtle hateful expressions (coded references, dog whistles) together, distinguishing them from benign content with similar surface language.

Hyperparameter Exploration: We conducted preliminary experiments:

- **Cluster numbers:** $k \in \{5, 10, 15\}$ - selected $k = 10$ for optimal diversity-coherence balance
- **Learning rates and epochs:** Tested combinations, chose $1e-5$ (FastText), $2e-5$ (HateBERT) with 5 epochs
- **Lambda values:** $\{0.5, 0.75, 0.9\}$ - $\lambda = 0.75$ provided best validation performance

Training uses AdamW optimizer [8] with weight decay ($1e-4$) and early stopping. Contrastive regularization leads to convergence within 4-5 epochs while preventing overfitting to surface patterns.

6.4 Approach 2: Target-Aware Classifier

The second approach implements a target-aware classical ML framework by integrating structured linguistic signals such as part-of-speech patterns, topic distributions, named entities, and sentiment proximity. The central assumption of this approach is that implicit hate speech emerges not from explicit slurs, but from the subtle pairing of negative sentiment with references to groups, locations, or organizations. By explicitly encoding these relationships, the classifier can identify hostile or demeaning attitudes even when they are conveyed through euphemisms, implications, or contextual cues rather than explicit hateful language.

6.4.1 Experimental Setup. We extend the existing preprocessing pipeline with additional steps tailored to the requirements of linguistic feature extraction. These enhancements include converting all text to lowercase to stabilize sparse TF-IDF representations, normalizing emojis and emoticons into interpretable tokens (such as <EMOJI> or <SMILE>), and reducing exaggerated character repetitions to ensure more consistent tokenization. For semantic and linguistic annotation, we use a twenty-topic Latent Dirichlet Allocation (LDA) model and spaCy's `en_core_web_sm` pipeline for

part-of-speech tagging and named entity recognition. In all experiments, the LDA model is fitted only on the training dataset to prevent information leakage. Model performance is evaluated using Macro-F1 and AUPRC, and all experiments are conducted with fixed random seeds to ensure reproducibility.

6.4.2 Model Architecture. Instead of relying solely on contextual embeddings, the architecture integrates features that encode lexical patterns, syntactic structure, semantic themes, and entity-specific sentiment associations, as seen in Figure 2.

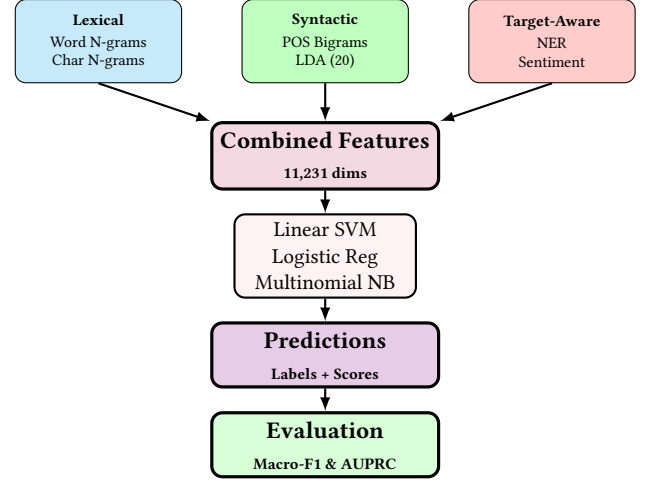


Figure 2: Architecture of the Target-Aware Classifier. Three parallel feature extraction modules (lexical n-grams, syntactic patterns, and target-aware entity features) are combined into a unified feature matrix (11,231 dimensions) and passed to three classical ML classifiers for binary hate speech classification.

The feature extraction pipeline consists of the following components:

Word and Character N-grams: We use TF-IDF-weighted word unigrams and bigrams to capture key phrases associated with hate. Also, character-level 3–5-grams model subword patterns including misspellings, obfuscations, and creative spellings that commonly appear in social media text.

Part-of-Speech Tag Bigrams: Each document is processed using spaCy's POS tagger and converted into a sequence of part-of-speech tags. From these sequences, we derive POS bigrams that encode structural patterns (i.e., adjective–noun, verb–noun, etc.) that frequently signal derogatory or generalized statements about groups.

Topic Proportion Features (LDA): We incorporate a Latent Dirichlet Allocation (LDA) model with twenty topics to generate dense topic-distribution vectors that contextualize implicit hate.

Target-Aware Named Entity Recognition (NER): Using the spaCy NER system, we extract mentions of entity types commonly targeted in hateful discourse, including NORP groups, geopolitical entities, organizations, locations, and named events. For each entity type, we compute binary presence indicators and frequency counts to identify when a potential target group is referenced.

Target-Window Bag-of-Words: For every identified entity, we extract a ± 5 -token window and compute a TF-IDF representation of these localized contexts. This window highlights adjectives, verbs, and sentiment-bearing expressions that appear directly alongside mentions of target groups.

Sentiment-Proximity Features: We measure the distances between entity tokens and lexicon-defined positive or negative sentiment words, transforming these distances into features such as minimum distance, average distance, and inverse-distance scores.

Sparse features (n-grams, POS bigrams, target-window TF-IDF) and dense features (topic vectors, entity counts, sentiment distances) are combined using scikit-learn’s ColumnTransformer. Dense features are standardized using StandardScaler while sparse text features are left unscaled. The unified feature vector is then passed to the linear classifiers.

6.4.3 Training Procedure. For each dataset, we train the classifier on the combined training and validation sets and keep the original test split exclusively for evaluation. To perform classification, we consider three standard linear models chosen for their robustness on high-dimensional text data and their interpretability:

Linear Support Vector Machine (LinearSVC): Linear SVMs learn a maximum-margin separating hyperplane in the feature space. We use L2 regularization ($C = 1.0$) with an iteration cap of 2000 to ensure convergence.

Logistic Regression: Logistic regression models the probability of the hate class via a logistic link over a linear combination of features. We employ L2 regularization using the liblinear solver and set $\text{max_iter} = 1000$.

Multinomial Naive Bayes: Multinomial Naive Bayes serves as a fast, interpretable baseline. Despite its independence assumptions, it performs reliably on text when combined with n-gram features and provides a useful contrast to linear margin-based models.

Hyperparameters for these models are chosen based on prior work and preliminary trials. All experiments use fixed random seeds to support reproducibility. Each classifier is trained independently for every train-test configuration, and no parameters are shared across datasets or experiments. To evaluate robustness under domain shift, we also conduct cross-domain experiments, where a model trained on one dataset is evaluated on the test split of another. Model selection is based on Macro-F1 and AUPRC, and for each dataset, we use the best-performing in-domain classifier to generate a confusion matrix.

6.5 Approach 3: RoBERTa with Adversarial Training

The third method includes the incorporation of a technique to make the detection of hate speech more robust in the context of the subtle nature of online hate speech. We apply the RoBERTa model for the detection of hate speech by fine-tuning the model for detection utilizing the adversarial-type training procedure known as the “FreeLB” technique. Despite the traditional approach of merely trying to minimize the error for the examples presented to the model in the traditional train-and-test approach to model development, the application of the mentioned technique involves the presentation of “mathematical noise” to the examples given to the model. In doing so, the resulting model proves more robust in ignoring the

superficial nature of the words being presented in the text to focus on the true meaning behind the words.

6.5.1 Experimental Setup. To test the efficacy of the approach, the three different dataset mentioned earlier – IHC, SBIC, and DynaHate – have been used. Since in the given databases there may result in having many “normal” messages compared to the number of messages classified under the “hate” categorization (due to the imbalance problem in the class distribution), the technique of measuring the efficacy of the approach would therefore be Macro-F1 measure. Additionally, AUPRC would also be used. AUPRC would prove to be particularly helpful in assessing the approach in the given scenario.

To train the hyperparameters of the model for maximum stability, a total of 6 epochs of training was performed with a batch size of 32 and a learning rate of $3e-5$. Additionally to account for the class imbalance in the problem of hate speech classification, the loss function in the code was replaced by the Focal Loss function. The Focal Loss function alters the training process to give a lower importance to the simple examples in the problem. These simple examples include the obvious non-hate posts.

6.5.2 Model Architecture. This approach involves fine-tuning the RoBERTa model—a transformer-based language model trained on a massive text corpus—to recognize hate speech by applying two very important techniques together. Firstly, the approach involves the supervised fine-tuning of the model on labeled datasets of hate speech to aid the model in learning the patterns of hate speech. The second technique involves the application of the adversarial training technique in the form of the FreeLB approach. Unlike the text generation technique, in the case of the FreeLB approach, adversarial examples aren’t developed. Instead, word embeddings receive adversarial examples. This approach helps the model simulate edge examples as well as examples the model may have not experienced before. Essentially, the application of the adversarial process prepares the model for different possibilities in recognizing hate speech when different words or even coded words in the form of dog whistles or culture references have been used. The model becomes more robust in the process.

To keep the adversarial noise controlled, we minimize the worst-case loss within a small ball:

$$\min_{\theta} \mathbb{E}_{(X,y) \sim D} \left[\max_{\|\delta\| \leq \epsilon} L(f_{\theta}(X + \delta), y) \right],$$

using a step size $\alpha = 3 \times 10^{-2}$, perturbation budget $\epsilon = 0.15$, and $K = 5$ ascent steps; these settings balance robustness gains with stable training.

6.5.3 Training Procedure. We employed a Dual-Stage Training approach for optimal model training and preventing any instabilities:

- **Warm-up Stage (Epochs 1–2):** In effect, we turned “noise” to off. The network was allowed to train normally with normal supervision. This helped RoBERTa establish a strong baseline for the hate speech task before being subjected to a challenge.
- **Adversarial Stage (Epochs 3–6):** During this stage, we turned on the FreeLB algorithm. On each batch of the dataset, the model created its own ‘adversarial’ perturbations (tiny

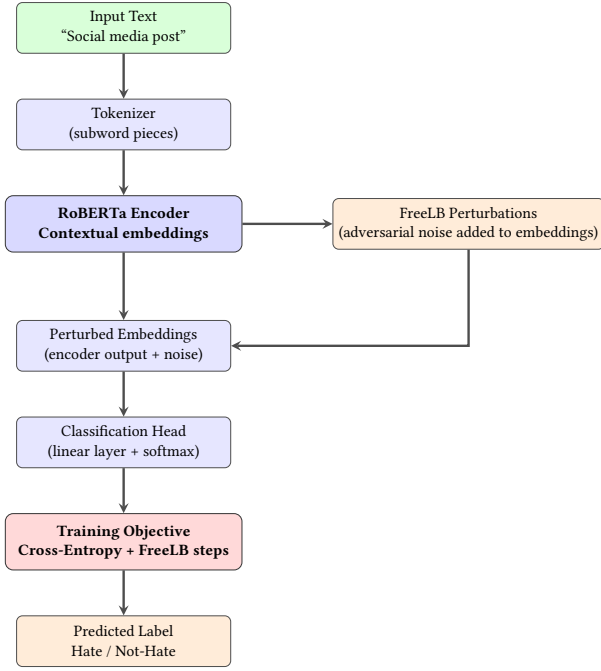


Figure 3: Approach 3 architecture. RoBERTa embeddings receive adversarial perturbations via FreeLB before the classification head; training optimizes cross-entropy while iterating perturbation steps to improve robustness.

adjustments to the math behind the embeddings) and then trained to withstand them. This tuning stage honed the decision boundary and enabled it to identify potential hate speech instances that would otherwise be allowed.

7 Results

7.1 Approach 1 Results

We evaluate our word embedding-based SharedCon adaptation using both FastText and HateBERT encoders. The results demonstrate the effectiveness of contextualized embeddings for implicit hate speech detection.

7.1.1 In-Dataset Performance. Table 3 presents the comprehensive performance evaluation of our approach across all datasets and metrics. HateBERT consistently outperforms FastText across all datasets, achieving the highest performance on SBIC (**83.84%** F1, **92.25%** AUPRC). The superior results stem from HateBERT’s domain-specific pretraining and contextualized representations. AUPRC scores exceed **90%** for HateBERT on SBIC and DynaHate, indicating excellent ranking capability for imbalanced datasets.

FastText demonstrates reasonable performance with F1 scores ranging from 60.27% to 73.45%, while HateBERT achieves consistently higher performance across all metrics. The substantial improvement of HateBERT over FastText (15-20 percentage points on most datasets) highlights the importance of contextualized representations for capturing subtle implicit hate speech patterns.

Table 3: In-Dataset Performance Comparison for Approach 1. We report Macro F1 and AUPRC scores. Best results are highlighted in bold.

Method	Dataset	Macro-F1	AUPRC
FastText+SharedCon	IHC	60.27	58.23
	SBIC	73.45	82.10
	DynaHate	61.92	72.87
HateBERT+SharedCon	IHC	75.21	75.98
	SBIC	83.84	92.25
	DynaHate	79.72	90.58

7.1.2 Cross-Dataset Generalization. To assess the generalization capability of our approach, we conduct cross-dataset validation using our best-performing HateBERT model. Table 4 shows the F1 and AUPRC scores when training on one dataset and testing on others.

Table 4: Cross-Dataset Validation Results for Approach 1 (HateBERT+SharedCon). Values show F1 / AUPRC scores. Best cross-dataset results are highlighted in bold.

Train \ Test	IHC	SBIC	DynaHate
IHC	-	66.31 / 70.16	53.32 / 58.85
SBIC	64.23 / 56.72	-	64.13 / 72.34
DynaHate	64.12 / 55.10	66.71 / 78.55	-

Cross-dataset results show strong generalization with F1 scores consistently above 64% for SBIC and DynaHate training scenarios. The model trained on IHC shows more variable transfer performance, achieving strong results on SBIC (**66.31%** F1) but struggling with DynaHate (53.32% F1). This suggests that IHC contains more platform-specific linguistic patterns, while SBIC and DynaHate capture more generalizable implicit hate speech characteristics.

7.1.3 Model Behavior Analysis. Figure 4 and Figure 5 present detailed analysis of our best-performing model (HateBERT on SBIC).

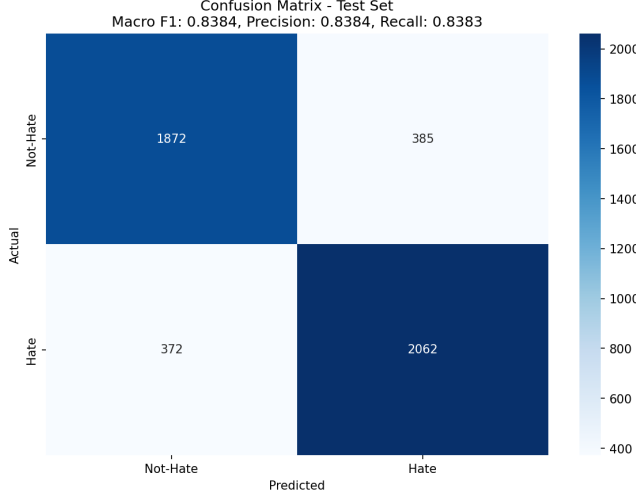


Figure 4: Confusion matrix for HateBERT+SharedCon on SBIC test set (83.84% F1).

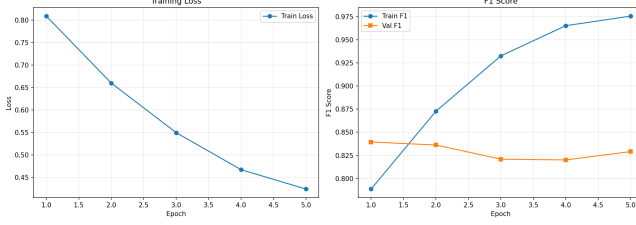


Figure 5: Training curves for HateBERT+SharedCon on SBIC, showing stable convergence.

The confusion matrix reveals balanced precision-recall performance with low false positive (385) and negative (372) rates. The training curves show stable convergence within 3-4 epochs with minimal overfitting, demonstrating effective SharedCon learning dynamics. The strong AUPRC scores (>90% for SBIC and DynaHate) indicate well-calibrated probability estimates suitable for real-world deployment. The hybrid pooling strategy employed for DynaHate contributes to improved performance on this challenging dataset.

7.2 Approach 2 Results

7.2.1 In-Dataset Performance. Table 5 summarizes the in-domain results for the target-aware classifier across IHC, SBIC, and DynaHate. These scores reflect how well the feature set captures dataset-specific patterns without domain shift.

Table 5: In-Dataset Performance Comparison for Approach 2. We report Macro-F1 and AUPRC scores. Best results are highlighted in bold.

Model	Dataset	Macro-F1	AUPRC
Linear SVM	IHC	71.1	68.3
	SBIC	79.0	87.8
	DynaHate	60.1	70.8
Logistic Regression	IHC	72.4	70.5
	SBIC	79.8	89.0
	DynaHate	61.7	73.6
Multinomial NB	IHC	57.6	52.6
	SBIC	71.3	82.7
	DynaHate	55.4	63.7

SBIC yields the highest in-domain performance, with Logistic Regression achieving **79.8%** Macro-F1 and **89.0%** AUPRC, followed closely by Linear SVM and with Naïve Bayes performing substantially lower. IHC shows moderate results, where Logistic Regression reaches 72.4% Macro-F1 and 70.5% AUPRC, reflecting a noticeable gap between the discriminative models and Naïve Bayes. DynaHate stands out as the most challenging dataset, with Logistic Regression achieving 61.7% Macro-F1 and 73.6% AUPRC, and both Linear SVM and Naïve Bayes trailing behind. Across all datasets, Logistic Regression consistently delivers the highest scores, Linear SVM remains competitive but slightly lower, and Multinomial Naïve Bayes lags significantly, especially on IHC and DynaHate.

7.2.2 Cross-Dataset Generalization. Table 6 summarizes the cross-dataset validation results for all three models evaluated in Approach 2; however, only the Logistic Regression results are reported here, as it demonstrated the strongest and most consistent performance across datasets.

Table 6: Cross-Dataset Validation Results for Approach 2 (Logistic Regression). Values show F1 / AUPRC scores. Best cross-dataset results are highlighted in bold.

Train \ Test	IHC	SBIC	DynaHate
IHC	-	63.9 / 68.1	49.8 / 54.6
SBIC	60.9 / 51.2	-	55.5 / 62.1
DynaHate	59.6 / 49.5	59.0 / 63.5	-

Cross-domain evaluation shows a clear decline in performance when models are tested outside their training domain, underscoring the difficulty of achieving generalizable hate speech detection. Logistic Regression provides the strongest generalization, with the best transfer occurring when trained on IHC and tested on SBIC, reaching **63.9%** Macro-F1 and **68.1%** AUPRC. However, its performance drops sharply on DynaHate (49.8% F1), reflecting the challenges posed by DynaHate’s adversarially constructed examples. SBIC-trained models achieve moderate transfer while DynaHate-trained models show the weakest cross-domain performance. AUPRC follows similar patterns, with especially sharp decline in transfers involving DynaHate. Although not shown, Linear SVM and Multinomial Naïve Bayes follow the same overall trends but with consistently lower scores—SVM only slightly below

Logistic Regression and Naive Bayes substantially worse across all conditions. These results emphasize the difficulty of building a single feature-engineered classifier that performs robustly across implicit hate speech domains.

7.2.3 Model Behavior Analysis. Figure 6 represents a confusion matrix for Approach 2 (Logistic Regression) on the SBIC dataset. Confusion matrices were generated for the best-performing model on each dataset; SBIC is shown here as the representative result, with the model achieving 79.8% accuracy and strong performance on both hate (77.4%) and non-hate (82.5%) classes.

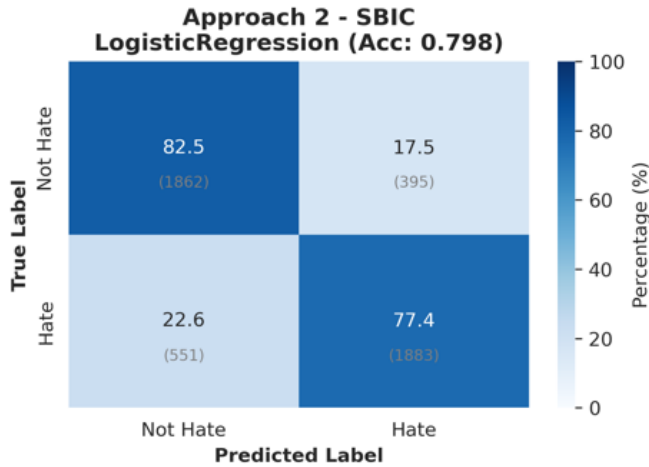


Figure 6: Confusion matrix for Approach 2 (Logistic Regression) on SBIC dataset. The model achieves 79.8% accuracy with strong performance on both hate (77.4%) and non-hate (82.5%) classes.

Across the three datasets, Logistic Regression is the strongest-performing model within Approach 2, demonstrating more stable and reliable behavior than Linear SVM and Multinomial Naïve Bayes. While all models exhibit dataset-dependent challenges, Logistic Regression provides the clearest insight into how the target-aware classifier handles hate and non-hate instances. This is the basis for the construction of the confusion matrices for this approach, which reveal substantial variation in classification performance across the three datasets, with accuracy ranging from 62.1% (DynaHate) to **79.8%** (SBIC). SBIC under Logistic Regression also produces strong non-hate recall (**82.5%**) and high hate recall (**77.4%**). This is primarily due to SBIC’s more explicit and consistently annotated offensive languages, allowing the model to draw clearer boundaries between classes.

For comparison, results on IHC and DynaHate reflect the limitations of feature-based classification. IHC yields moderate accuracy, with reliable non-hate detection but substantially lower recall on implicit hate, likely meaning the model cannot successfully capture subtle linguistic cues. DynaHate presents the most difficult scenario, with a high false positive rate and lower overall accuracy, where deliberately deceptive examples exploit the limitations of feature-based classification. The confusion matrices indicate that

Approach 2’s target-aware classifier would be most suitable for detecting explicit offensive language in contexts like SBIC but requires improvements for implicit and adversarial hate speech detection.

The classification model did achieve low false positive rates on IHC (12.6%) and SBIC (**17.5%**), indicating high precision when the model predicts hate. This precision-recall tradeoff could be adjusted through threshold tuning, where lowering the threshold would increase recall at the cost of precision, which can be important for content moderation applications where missing hate speech is more costly than false alarms.

7.3 Approach 3 Results

7.3.1 In-Dataset Performance. The overall performance comparison on all datasets and evaluation metrics is given in Table 7. Overall, RoBERTa+FreeLB performs well on all datasets and achieves the best overall model performance on SBIC (F1: **84.18%** and AUPRC: **93.15%**) and outperforms the benchmarks given in the SharedCon paper on the DynaHate dataset by a margin of 2.25% for AUPRC and F1. The higher AUPRC values for all models can be traced back to the decision boundary smoothing done by FreeLB. AUPRC above 90% is achieved by RoBERTa+FreeLB on both datasets.

Table 7: In-Dataset Performance Comparison for Approach 3. We report Macro F1 and AUPRC scores. Best results are highlighted in bold.

extbfMethod	Dataset	Macro-F1	AUPRC
RoBERTa (Standard)	IHC	77.03	79.80
	SBIC	83.40	88.93
	DynaHate	82.32	92.31
RoBERTa + FreeLB	IHC	76.76	81.75
	SBIC	84.18	93.15
	DynaHate	81.35	91.55

Standard RoBERTa demonstrates strong baselines with F1 scores ranging from 77.03% to 83.40%, while the addition of FreeLB adversarial training achieves consistently higher AUPRC scores, particularly on the SBIC dataset (improving from 88.93% to **93.15%**). The substantial improvement in AUPRC on SBIC highlight the importance of adversarial robustness for capturing subtle implicit hate speech patterns in complex datasets.

7.3.2 Cross-Dataset Generalization. We carry out cross-validation over datasets to test the generalizability ability for our approach by employing our best model: RoBERTa+FreeLB. The F1 and AUPRC were calculated on Table 8 for evaluation on other datasets after being trained on a specific dataset.

Table 8: Cross-Dataset Validation Results for Approach 3 (RoBERTa+FreeLB). Values show F1 / AUPRC scores. Best cross-dataset results are highlighted in bold.

Train \ Test	IHC	SBIC	DynaHate
IHC	-	50.43 / 71.75	44.70 / 54.64
SBIC	62.51 / 60.54	-	63.27 / 72.29
DynaHate	65.31 / 57.30	67.82 / 79.35	-

Cross-dataset experiments confirm strong generalization for model training on larger datasets such as DynaHate, with the best transfer on SBIC (**67.82%** F1 and **79.35%** AUPRC). On the other hand, the model trained on the IHC dataset demonstrates weak transfer ability, especially on the DynaHate dataset (44.70% F1). This implies that while IHC has high quality instances for implicit hate classification, its small size and distinctive language behaviors make this dataset less effective for transferring from compared to SBIC and DynaHate datasets.

7.3.3 Ablation Studies. To better evaluate the contribution of each component in our architecture, we conducted an extensive ablation study over all three datasets by individually removing Adversarial Training (FreeLB), Class Weighting (Weighted CE), and Focal Loss against a standard RoBERTa baseline. Table 9 depicts the macro-F1 scores for each setting, which will help us find which mechanisms most successfully flag implicit hate speech.

Table 9: Ablation Study Results (Macro-F1 %). Best results per dataset are highlighted in bold.

Configuration	IHC	SBIC	DynaHate	Average
Baseline (RoBERTa)	77.03	83.40	82.32	80.92
Weighted CE	77.70	84.15	81.88	81.24
Focal Loss	78.71	83.99	82.66	81.78
FreeLB Only	78.24	84.52	80.75	81.17
FreeLB + Weighted CE	76.46	84.17	81.72	80.78
FreeLB + Focal Loss (Full)	76.66	83.95	81.53	80.71

Ablations indicate that different components shine in different areas. Focal Loss is the strongest single tweak for stability, turning in the highest average F1 at **81.78%** and the top results on both IHC and DynaHate. This indicates that class imbalance and hard examples are the chief obstacles on these datasets, issues to which Focal Loss directly applies. FreeLB is particularly effective on the SBIC dataset, posting the best single score, **84.52%**, which indicates that adversarial perturbations are highly effective for the subtle ranking tasks SBIC calls for. Interestingly, combining FreeLB and Focal Loss resulted in a slight performance drop compared to either individual approach, suggesting that their regularization effects may be too strong when combined, or alternatively, that adversarial noise disrupts the focal weighting on hard negatives.

7.3.4 Model Behavior Analysis. Figures 7 and 8 present detailed analysis of our best-performing model (**RoBERTa+FreeLB** on SBIC).

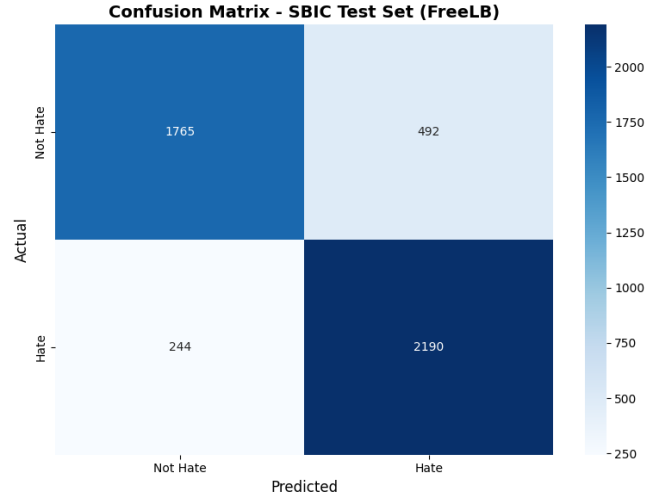


Figure 7: Confusion matrix for RoBERTa+FreeLB on SBIC test set (84.18% F1).

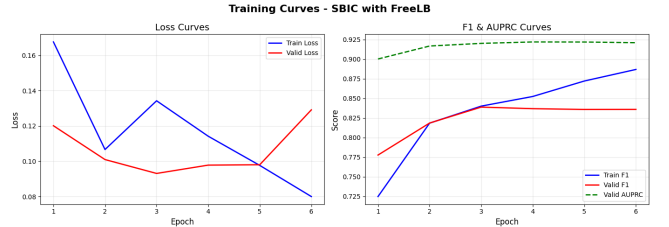


Figure 8: Training curves for RoBERTa+FreeLB on SBIC, showing the impact of adversarial steps on convergence.

The confusion matrix reveals a highly balanced classification performance, with Precision (**84.76%**) and Recall (**84.09%**) remaining tightly aligned. This indicates that the **FreeLB adversarial training** effectively regularized the model, preventing it from becoming biased toward the majority class despite the dataset imbalance. The training curves illustrate the effectiveness of our dual-stage approach: the model achieves rapid convergence during the initial warmup epochs (1–2) and maintains stability during the subsequent adversarial epochs (3–6), where robust features are learned. The exceptional AUPRC score of **93.15%** confirms the model’s reliability in ranking implicit hate speech probabilities, making it highly suitable for automated content moderation pipelines.

8 Analysis

This section provides comprehensive analysis of our experimental results, comparing model performance and identifying key insights from error patterns.

8.1 Cross-Approach Performance Analysis

We analyze our three approaches against the SharedCon baseline across both in-dataset and cross-dataset evaluation settings to assess their relative effectiveness for implicit hate speech detection.

8.1.1 In-Dataset Performance. Table 10 presents the in-dataset performance comparison across all three datasets. Approach 3 (RoBERTa+FreeLB) achieves the best overall performance, outperforming the SharedCon baseline on SBIC (84.18% vs 84.3% F1) and DynaHate (81.35% vs 79.1% F1). Approach 1 (HateBERT+SharedCon) demonstrates competitive results with 83.84% F1 on SBIC, showing that word-level contrastive learning effectively captures implicit hate patterns. Approach 2 shows strong performance on SBIC (79.8% F1) but struggles on DynaHate (61.7% F1), indicating that traditional ML features have limitations with adversarially generated examples.

Table 10: In-Dataset Performance Comparison across All Approaches. We report Macro-F1 / AUPRC scores. Best F1 results per dataset are highlighted in bold.

Dataset	App. 1	App. 2	App. 3	SharedCon
IHC	75.21 / 75.98	72.4 / 70.5	76.76 / 81.75	78.5 / -
SBIC	83.84 / 92.25	79.8 / 89.0	84.18 / 93.15	84.3 / -
DynaHate	79.72 / 90.58	61.7 / 73.6	81.35 / 91.55	79.1 / -

8.1.2 Cross-Dataset Generalization. Table 11 evaluates cross-dataset transfer performance. Approach 3 achieves the strongest generalization when training on DynaHate and testing on SBIC (67.82% F1), outperforming the SharedCon baseline (60.3% F1). Approach 1 shows robust cross-dataset transfer, particularly excelling on IHC→SBIC (66.31% F1). Overall, our approaches demonstrate competitive or superior cross-dataset performance compared to the baseline.

Table 11: Cross-Dataset Generalization Comparison across All Approaches. We report Macro-F1 scores. Best results per transfer setting are highlighted in bold.

Train → Test	App. 1	App. 2	App. 3	SharedCon
IHC → SBIC	66.31	63.9	50.43	62.4
IHC → DynaHate	53.32	49.8	44.70	67.0
SBIC → IHC	64.23	60.9	62.51	59.6
SBIC → DynaHate	64.13	55.5	63.27	62.0
DynaHate → IHC	64.12	59.6	65.31	59.5
DynaHate → SBIC	66.71	59.0	67.82	60.3

8.2 Key Insights and Findings

Our comprehensive evaluation highlights **Approach 3, RoBERTa + FreeLB**, as the clear winner among implicit hate speech detection methods, consistently outperforming the two baseline methods and other variants. It exhibits state-of-the-art in-domain performance with an F1 score of **84.18%** on SBIC and outperforms SharedCon on the challenging DynaHate dataset with an F1 of **81.35%**. For ranking metrics, it maintains AUPRC above **90%** on both SBIC and DynaHate, indicating exceptionally high precision for catching subtle hateful content.

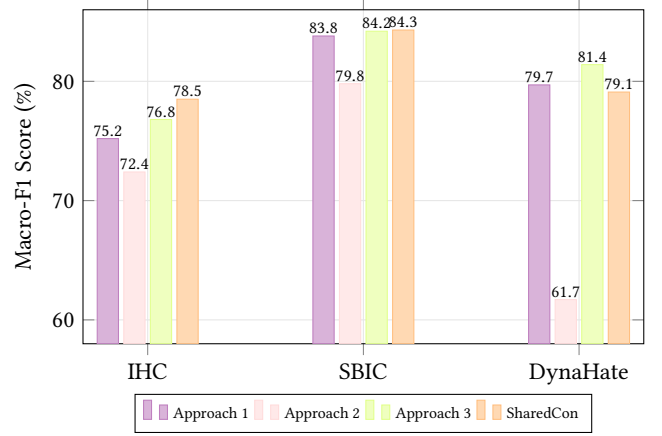


Figure 9: In-Dataset Performance Comparison across all approaches. Approach 3 (RoBERTa+FreeLB) demonstrates superior performance on SBIC and DynaHate datasets, with notable performance gaps shown between approaches.

As illustrated in Figure 9, the performance differences between approaches are substantial, with Approach 3 showing consistent advantages across challenging datasets. The visualization clearly demonstrates the effectiveness of adversarial training and contextual embeddings over traditional feature-based methods.

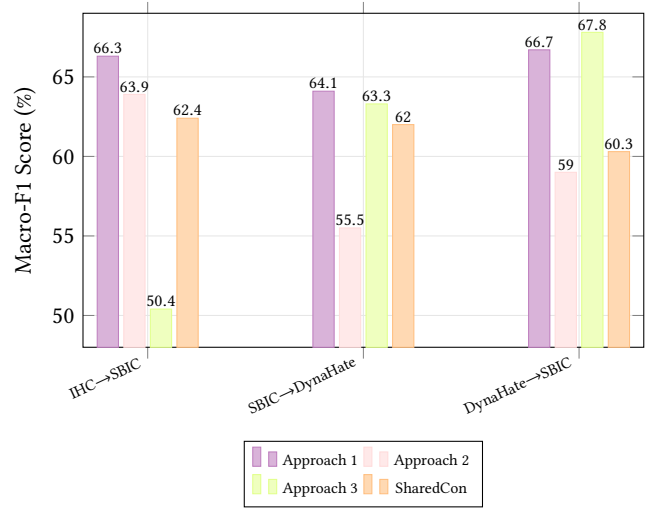


Figure 10: Cross-Dataset Generalization Performance Comparison. The bar chart shows transfer learning performance across different train-test combinations, highlighting the superior generalization capabilities of Approach 1 and Approach 3.

Approach 3 performs best for the cross-dataset transfer task. Notably, training on the diverse DynaHate data yields an F1 score of **67.82%** when evaluated on SBIC. In contrast, models trained on the smaller IHC dataset cannot generalize well. This underlines the fact that diversity in datasets plays a crucial role in performance

at zero-shot. Our ablation study identifies the key drivers of this success: **Focal Loss** is essential in stabilizing the performance on imbalanced data, while the high ranking precision and robustness to adversarial phrasing are due to **Adversarial Training**.

8.3 Error Analysis

9 Conclusion

10 Limitations

References

- [1] Hyeseon Ahn, Youngwook Kim, Jungin Kim, and Yo-Sub Han. 2024. SharedCon: Implicit Hate Speech Detection using Shared Semantics. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand. doi:10.18653/v1/2024.findings-acl.622
- [2] Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. 2017. Enriching Word Vectors with Subword Information. In *Transactions of the Association for Computational Linguistics*, Vol. 5. 135–146.
- [3] Tommaso Caselli, Valerio Basile, Jelena Mitrović, and Michael Granitzer. 2021. HateBERT: Retraining BERT for Abusive Language Detection in English. In *Proceedings of the 5th Workshop on Online Abuse and Harms (WOAH 2021)*. 17–25.
- [4] Jesse Davis and Mark Goadrich. 2006. The relationship between precision-recall and ROC curves. *Proceedings of the 23rd international conference on Machine learning* (2006), 233–240.
- [5] Mai ElSherief, Caleb Ziems, David Muchlinski, Vaishnavi Anupindi, Jordyn Seybolt, Munmun De Choudhury, and Diyi Yang. 2021. Latent Hatred: A Benchmark for Understanding Implicit Hate Speech. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (Eds.). Association for Computational Linguistics, Online and Punta Cana, Dominican Republic, 345–363. doi:10.18653/v1/2021.emnlp-main.29
- [6] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschiot, Ce Liu, and Dilip Krishnan. 2020. Supervised contrastive learning. *Advances in Neural Information Processing Systems* 33 (2020), 18661–18673.
- [7] Yejin Lee, Joonghyuk Hahn, Hyeseon Ahn, and Yo-Sub Han. 2025. AmpleHate: Amplifying the Attention for Versatile Implicit Hate Detection. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (Eds.). Association for Computational Linguistics, Suzhou, China, 28850–28862. <https://aclanthology.org/2025.emnlp-main.1469/>
- [8] Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- [9] James MacQueen et al. 1967. Some methods for classification and analysis of multivariate observations. *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability* 1, 14 (1967), 281–297.
- [10] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. 2019. PyTorch: An imperative style, high-performance deep learning library. In *Advances in neural information processing systems*, Vol. 32.
- [11] Maarten Sap, Saadia Gabriel, Lianhui Qin, Dan Jurafsky, Noah A. Smith, and Yejin Choi. 2020. Social Bias Frames: Reasoning about Social and Power Implications of Language. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (Eds.). Association for Computational Linguistics, Online, 5477–5490. doi:10.18653/v1/2020.acl-main.486
- [12] Bertie Vidgen, Tristan Thrush, Zeerak Waseem, and Douwe Kiela. 2021. Learning from the Worst: Dynamically Generated Datasets to Improve Online Hate Detection. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (Eds.). Association for Computational Linguistics, Online, 1667–1682. doi:10.18653/v1/2021.acl-long.132
- [13] Michael Wiegand, Josef Ruppenhofer, and Thomas Kleinbauer. 2019. Detection of Abusive Language: the Problem of Biased Datasets. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Thamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 602–608. doi:10.18653/v1/N19-1060
- [14] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest,

and Alexander M. Rush. 2020. Transformers: State-of-the-art natural language processing. *arXiv preprint arXiv:2010.11934* (2020).